



POLITECNICO
MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

Analytical Report of the First Assignment

AERONAUTICAL ENGINEERING - STRUCTURAL DYNAMICS AND AEROELASTICITY COURSE

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1. Introduction

The assigned problem (graphically shown in Fig.1) consists of a straight wing of span b and chord c , clamped at the root ($y = 0$) and free at the tip ($y = b$). The wing is characterized by constant bending stiffness EJ and torsional stiffness GJ . The Elastic Axis (EA) is located at the mid-chord, while the Aerodynamic Center (AC) is positioned at a distance $e = c/4$ forward of the EA. To enhance the structural stability and modify the aeroelastic response, a strut, with axial stiffness EA_{strut} , is integrated into the system. The strut connects the wind tunnel wall (Point A) to the wing midspan ($y = b/2$, Point B). The connection at Point B is defined by a chordwise offset d relative to the Elastic Axis. The strut forms an angle γ with the spanwise axis of the wing. The wing is equipped with an aileron of span b_a , located at the outboard section. The goal of this assignment is to investigate the interaction between structural elasticity, concentrated constraints (the strut) and aerodynamic loads to evaluate critical phenomena such as divergence and control reversal.

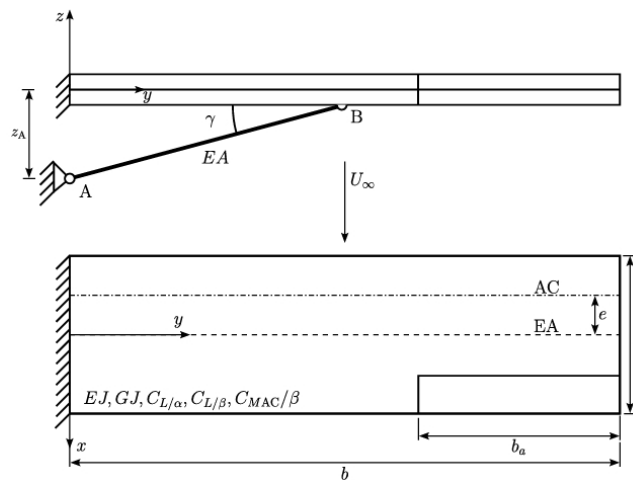


Figure 1: Set-up of the problem

2. Analytical Model Development

2.1. Degrees of Freedom (DoF) and Kinematics

First of all, it is necessary to identify the Degrees of Freedom (DoF) of the system. The state of the beam can be described by two displacement fields:

1. $w(y)$: vertical displacement of the Elastic Axis;
2. $\theta(y)$: torsional rotation of the cross-section about the Elastic Axis.

The displacement of a generic point $P(x, y)$ on the wing chord can be written as

$$u(x, y) = w(y) - x\theta(y) \quad (1)$$

where x is the distance between point P and the Elastic Axis.

In a complete formulation, the local angle of attack of the section is given by

$$\alpha(y) = \theta(y) + w'(y),$$

where the bending derivative $w'(y)$ contributes to the local aerodynamic incidence. In the present model, the contribution of $w'(y)$ is neglected, assuming small bending rotations compared to the torsional rotation. The aerodynamic loads are assumed to depend only on the torsional rotation $\theta(y)$.

2.2. Strut Kinematics

The strut is connected to the wing at midspan ($y_B = b/2$) at a specific chordwise location x_B . The vertical displacement of the attachment point B , denoted as u_B , is:

$$u_B = w\left(\frac{b}{2}\right) - x_B\theta\left(\frac{b}{2}\right). \quad (2)$$

The strut will elongate axially and so its vertical displacement component will be:

$$\delta_{\text{strut}} = u_B \sin(\gamma). \quad (3)$$

2.3. Principle of Virtual Works (PVW)

The kinematics of the problem can be summarized as the displacements of the Aerodynamic Center of the wing and the displacement of the attachment point B:

$$u_{AC} = w(y) + e\theta(y) \quad (4a)$$

$$u_B = w\left(\frac{b}{2}\right) - x_B\theta\left(\frac{b}{2}\right) \quad (4b)$$

where $e = c/4$ is the distance between the Aerodynamic Center (AC) and the Elastic Axis (EA).

Once the kinematics have been defined, the model development can proceed by writing the Principle of Virtual Work (PVW), stating that $0 = \delta W$ or, more precisely $\delta W_{\text{int}} = \delta W_{\text{ext}}$. δW is the total virtual work of the system and δW_{int} and δW_{ext} are respectively the internal and external virtual works.

The internal virtual work represents the elastic energy stored in the wing and the strut. Therefore, δW_{int} is the sum of the contributions from the wing's flexibility (bending and torsion) and the axial deformation of the strut:

$$\delta W_{\text{int}} = \int_0^b EJw''(y)\delta w''(y) dy + \int_0^b GJ\theta'(y)\delta\theta'(y) dy + \delta W_{\text{strut}} \quad (5)$$

The strut contribution acts as a concentrated spring at the midspan station $y_B = b/2$. Given the axial stiffness EA and the angle γ , the equivalent vertical stiffness K_B at point B is derived from the strut's kinematics:

$$\delta W_{\text{strut}} = K_B u_B \delta u_B, \quad \text{with} \quad K_B = \frac{EA}{L_{\text{strut}}} \sin^2(\gamma) \quad (6)$$

where $L_{\text{strut}} = \frac{b/2}{\cos(\gamma)}$ is the length of the strut.

The external work instead is generated by the distributed lift $L(y)$ and aerodynamic moment $M_{EA}(y)$ along the span:

$$\delta W_{\text{ext}} = \int_0^b L(y) \delta w(y) dy + \int_0^b M_{EA}(y) \delta \theta(y) dy \quad (7)$$

According to the strip theory and the provided coefficients, the aerodynamic forces are defined as:

$$L(y) = qcC_{L/\alpha}\theta(y) + qcC_{L/\beta}\beta(y) \quad (8a)$$

$$M_{EA}(y) = eL(y) + qc^2C_{MAC/\beta}\beta(y) \quad (8b)$$

2.4. Ritz-Galerkin Discretization

To solve the equations numerically, we apply the Ritz-Galerkin method. This approach approximates the continuous displacement fields $w(y)$ and $\theta(y)$ using a finite linear combination of trial functions (shape functions).

The trial functions must be kinematically admissible, meaning they must comply with the boundary conditions (b.c.): the wing root ($y = 0$) is clamped to the wind tunnel wall. The b.c. are:

- $w(0) = 0$ and $w'(0) = 0$ for bending;
- $\theta(0) = 0$ for torsion.

The approximations are defined as:

$$w(y) \approx \sum_{i=1}^{N_w} q_{wi} \phi_i(y) \quad (9a)$$

$$\theta(y) \approx \sum_{j=1}^{N_\theta} q_{\theta j} \psi_j(y) \quad (9b)$$

where $\phi_i(y)$ and $\psi_j(y)$ are the shape functions relative to bending and torsion respectively, while the coefficients q_{wi} and $q_{\theta j}$ are the unknowns of the problem. A suitable choice for the shape functions, which satisfies the boundary conditions, is the set of simple polynomials:

- Bending functions: $\phi_i(y) = \left(\frac{y}{b}\right)^{i+1}$ for $i = 1, \dots, N_w$
- Torsional functions: $\psi_j(y) = \left(\frac{y}{b}\right)^j$ for $j = 1, \dots, N_\theta$

Applying the new $w(y)$ and $\theta(y)$ expressions to the kinematics system (4) and substituting it in the PVW expression the following algebraic system is obtained:

$$(\mathbf{K}_s - q\mathbf{K}_a) \{\mathbf{q}\} = \{\mathbf{F}_{\text{ail}}\} \quad (10)$$

where $\mathbf{K}_s$ is the structural stiffness matrix obtained by solving

$$K_{s,ij} = \int_0^b EJ\phi_i''(y)\phi_j''(y) dy + \int_0^b GJ\psi_i'(y)\psi_j'(y) dy + K_B\mathbf{N}_B^T\mathbf{N}_B. \quad (11)$$

The aerodynamic stiffness matrix $\mathbf{K}_a$ is obtained solving

$$K_{a,ij} = \int_0^b cC_{L/\alpha}\psi_j(y)\phi_i(y) dy + \int_0^b ecC_{L/\alpha}\psi_j(y)\psi_i(y) dy. \quad (12)$$

The Generalized Aileron Force is obtained developing

$$F_{\text{ail},i} = \int_{b-b_a}^b cC_{L/\beta}\beta_0\phi_i(y) dy + \int_{b-b_a}^b (ecC_{L/\beta} + c^2C_{MAC/\beta})\beta_0\psi_i(y) dy \quad (13)$$

where β_0 is the deflection of the aileron.

2.5. Symbolic Solution

To simplify the symbolic derivation, a single degree of freedom is assumed for both displacement fields ($N_w = N_\theta = 1$). The lowest-order polynomials satisfying the geometric boundary conditions at the root ($y = 0$) are selected:

- bending shape function: $\phi_1(y) = \left(\frac{y}{b}\right)^2$;
- torsional shape function: $\psi_1(y) = \frac{y}{b}$.

By substituting these functions into the Principle of Virtual Works, the terms for the stiffness matrix $K_{s,ij}$ and the aerodynamic matrix $K_{a,ij}$ are derived.

The structural stiffness elements are determined as follows:

$$K_{s,11} = \int_0^b EJ(\phi_1'')^2 dy + K_B \phi_1(b/2)^2 = \frac{4EJ}{b^3} + \frac{K_B}{16} \quad (14)$$

$$K_{s,22} = \int_0^b GJ(\psi_1')^2 dy + K_B(x_B \psi_1(b/2))^2 = \frac{GJ}{b} + \frac{K_B x_B^2}{4} \quad (15)$$

$$K_{s,12} = K_{s,21} = K_B \phi_1(b/2) x_B \psi_1(b/2) = \frac{K_B x_B}{8} \quad (16)$$

The aerodynamic stiffness elements are:

$$K_{a,12} = \int_0^b cC_{L/\alpha} \psi_1 \phi_1 dy = \frac{cC_{L\alpha} b}{4} \quad (17)$$

$$K_{a,22} = \int_0^b ecC_{L/\alpha} \psi_1^2 dy = \frac{ecC_{L\alpha} b}{3} \quad (18)$$

and the components of the $\mathbf{F}_{\text{ail}}$ vector are:

$$F_{\text{ail},1} = \int_{b-b_a}^b qcC_{L/\beta} \beta_0 \left(\frac{y}{b}\right)^2 dy = qcC_{L/\beta} \beta_0 \left[\frac{y^3}{3b^2} \right]_{b-b_a}^b = \frac{qcC_{L/\beta} \beta_0}{3b^2} [b^3 - (b-b_a)^3] \quad (19)$$

$$\begin{aligned} F_{\text{ail},2} &= \int_{b-b_a}^b q(ceC_{L/\beta} + c^2 C_{MAC/\beta}) \beta_0 \left(\frac{y}{b}\right) dy \\ &= q(ceC_{L/\beta} + c^2 C_{MAC/\beta}) \beta_0 \left[\frac{y^2}{2b} \right]_{b-b_a}^b \\ &= \frac{q(ceC_{L/\beta} + c^2 C_{MAC/\beta}) \beta_0}{2b} [b^2 - (b-b_a)^2]. \end{aligned} \quad (20)$$

The complete algebraic system in matrix form becomes:

$$\begin{pmatrix} K_{s,11} & K_{s,12} \\ K_{s,21} & K_{s,22} \end{pmatrix} - q \begin{pmatrix} 0 & K_{a,12} \\ 0 & K_{a,22} \end{pmatrix} \begin{Bmatrix} q_{w1} \\ q_{\theta 1} \end{Bmatrix} = \begin{Bmatrix} F_{\text{ail},1} \\ F_{\text{ail},2} \end{Bmatrix}.$$

2.6. Divergence and Control Reversal Analysis

2.6.1 Divergence Dynamic Pressure q_D

Static aeroelastic divergence occurs when the structural stiffness is no longer able to counteract the increasing aerodynamic loads, leading to the failure of the structure. Mathematically, this corresponds to the condition where the determinant of the system's matrix vanishes, meaning the system can have a non-trivial solution even with zero external forcing ($\mathbf{F}_{\text{ail}} = 0$):

$$\det(\mathbf{K}_s - q\mathbf{K}_a) = 0. \quad (21)$$

By expanding the determinant for the 2×2 case:

$$K_{s,11}(K_{s,22} - qK_{a,22}) - (K_{s,12} - qK_{a,12})K_{s,21} = 0. \quad (22)$$

By solving this linear equation for q , the symbolic expression for the divergence dynamic pressure is obtained:

$$q_D = \frac{K_{s,11}K_{s,22} - K_{s,12}^2}{K_{s,11}K_{a,22} - K_{s,21}K_{a,12}}. \quad (23)$$

This formula highlights the influence of the strut: the term K_{s12} (which depends on x_B) couples bending and torsion, potentially increasing or decreasing the divergence threshold depending on the chordwise position of the attachment point. In fact if $x_B > 0$ (the attachment point is after the EA), the strut is opposing to the aerodynamic nose-up twist. This mechanism is expected to significantly delay divergence compared to the clean wing configuration.

2.6.2 Control Reversal Dynamic Pressure q_R

Control reversal occurs when the lift produced by the aileron deflection becomes null rendering the control surface useless.

Prior to determining the expression for q_R , the Control Effectiveness E_C is introduced. It is defined as the ratio between the lift produced by the flexible wing (L_{flex}) and the lift that would be produced if the wing was perfectly rigid (L_{rig}):

$$E_C(q) = \frac{L_{\text{flex}}(q)}{L_{\text{rig}}(q)}. \quad (24)$$

The rigid lift L_R depends only on the aileron deflection β_0 :

$$L_{\text{rig}} = \int_{b-b_a}^b qcC_{L/\beta}\beta_0 dy = qcC_{L/\beta}\beta_0 b_a. \quad (25)$$

The flexible lift L_{flex} includes the additional lift (or loss thereof) caused by the elastic twist $\theta(y)$ induced by the aileron's aerodynamic moment:

$$L_{\text{flex}} = L_{\text{rig}} + \int_0^b qcC_{L/\alpha}\theta(y)dy. \quad (26)$$

In our 2×2 symbolic model, E_C can be expressed by solving the system for $q_{\theta 1}$ and substituting it into the lift equation:

$$E_C(q) = 1 + \frac{qcC_{L/\alpha} \int_0^b \psi_1(y)dy}{L_{\text{rig}}} q_{\theta 1}(q). \quad (27)$$

The Control Reversal dynamic pressure, q_R , is formally defined as the pressure at which the control effectiveness drops to zero:

$$E_C(q_R) = 0 \implies L_{\text{flex}}(q_R) = 0. \quad (28)$$

2.7. Comparison of Deformed Shapes

In this section the kinematic impact of the strut is analysed comparing the deformed shape of the wing with and without the strut. To do so, it is necessary to compare the two displacements $w(y)$ and $\theta(y)$ computing the relative shape functions coefficients $q_{w,i}$ and $q_{\theta,i}$. The coefficient vector $\mathbf{q}$ is obtained solving the system expressed in the equation (10).

2.8. Control Reversal Analysis

The influence that the geometry of the strut has on the reversal dynamic pressure q_R was investigated through the evaluation of it as function of the chordwise offset x_B . What is expected from the numerical results is that as the attachment point B moves toward the trailing edge ($x_B > 0$), the strut provides a restorative moment that opposes the aileron-induced elastic twist. This effect shifts the reversal condition to higher dynamic pressure or it may remove this condition from the aircraft operational envelope.

3. Numerical Strategy and Convergence

While the 2×2 symbolic model provides fundamental physical insights, it relies on low-order polynomials that may not fully capture the local curvature discontinuity at the midspan station ($y = b/2$) caused by the concentrated strut force. In the numerical results report of this study, the number of shape functions N_w and N_θ will be increased. A convergence analysis will be performed to ensure that the divergence pressure q_D reaches a stable value, providing a reliable baseline for the subsequent structural optimization.